David Mukuruva

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Education

Dartmouth College — Hanover, NH
B.A., Double Major in Mathematics and Biomedical Engineering — *Expected June 2027*

Thayer School of Engineering — Hanover, NH
Bachelor & Master of Engineering in Biomedical Engineering — *June 2028*

Coursework: Probability (Honors), Real Analysis (Honors), Linear Algebra (Honors, proof-based), Complex Analysis (graduate), Combinatorial Game Theory, Differential Equations

Research Experience

FitzLab, Thayer School of Engineering — 2026 – Present
Undergraduate Researcher, Quantum Control — *Hanover, NH*

- Ran a 40-run test on Dartmouth's research supercomputer and found 2 of 8 cases frozen at a do-nothing baseline that the reported averages had hidden
- Removed one filtering step and a "clean" result's spread doubled, showing the filter drove the effect rather than the control method
- Reproduced the lab's benchmark behavior but not its published figure, and built the open-source tool the group now runs its tests on (public repo)

Blaschke Ellipses – Senior Honors Thesis (Mathematics) — June 2025 – Present
Researcher – Advisors: Prof. D. Williams, Prof. Y. Zeytuncu — *Hanover, NH / Remote*

- Confirmed both thesis problems were still unsolved by tracing every citation of the field's main survey through 2025
- Building the first odd-order ($n = 5$) examples of a construction previously known only for even cases, and testing whether a related family of curves is always an ellipse
- Co-authored a paper presented at the national Joint Mathematics Meetings in 2026, writing its computations in Python and Sage

Goods Lab, Thayer School of Engineering — Jan 2024 – Sep 2024
Research Assistant (Computational) — *Hanover, NH*

- Analyzed single-cell RNA data from cancer patients in R and mapped 16 cell types to surface candidate biomarkers
- Ran cell-cell communication analysis to map how tumor and immune cell populations signal to one another
- Built reproducible clustering pipelines in R and cross-referenced cell markers against patient metadata

Selected Projects

Studium | *local-first AI study tool — TypeScript, IndexedDB, OS Keychain* — 2026 – Present

- Built a study-drill module and a local-first data path (credentials in the OS keychain, data on-device); wrote the SSRF and body-cap tests that gate the network layer
- Wired the app to multiple AI providers through user-supplied keys, generating quizzes, flashcards, and concept maps from uploaded notes

Aleph-Nought | *Next.js (TypeScript), FastAPI, PostgreSQL* — 2025 – Present

- Built a symbolic engine that checks free-form math answers, inside a production platform serving six Dartmouth math and physics courses
- Automated the content pipeline with Manim, FFmpeg, Whisper, and Plotly, and shipped the stack on Vercel, Render, and Cloudflare R2

Skills & Interests

Technical: Python (NumPy/SciPy), Sage, Lean, R, C; SLURM/HPC, Git
Teaching: Lead Physics Tutor & TA, Dartmouth Emerging Engineers (10 weekly labs, 40+ students)
Interests: chess, competitive archery, combinatorial game theory